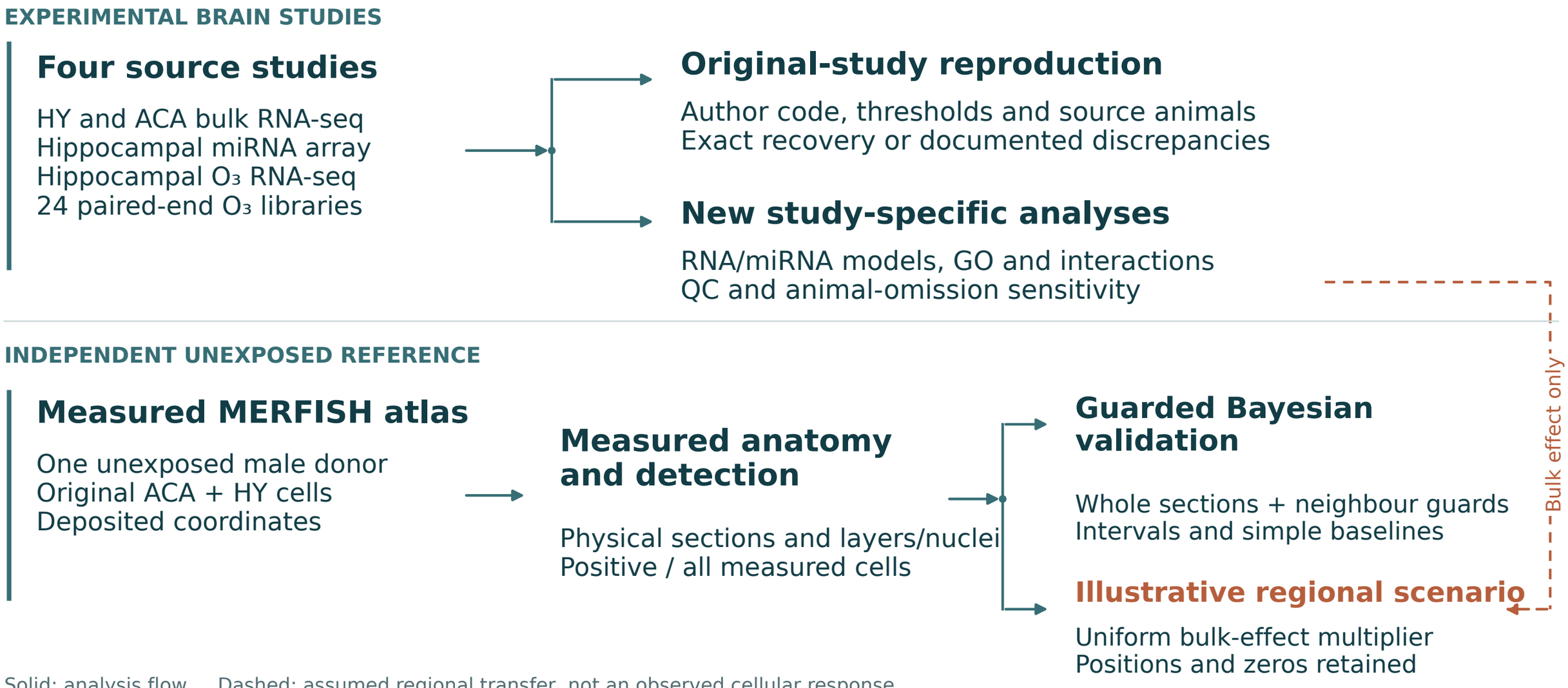
EXPERIMENTAL BRAIN STUDIES
Four source studies
HY and ACA bulk RNA-seq
Hippocampal miRNA array
Hippocampal O₃ RNA-seq
24 paired-end O₃ libraries
Original-study reproduction
Author code, thresholds and source animals
Exact recovery or documented discrepancies
New study-specific analyses
RNA/miRNA models, GO and interactions
QC and animal-omission sensitivity
INDEPENDENT UNEXPOSED REFERENCE
Measured MERFISH atlas
One unexposed male donor
Original ACA + HY cells
Deposited coordinates
Measured anatomy and detection
Physical sections and layers/nuclei
Positive / all measured cells
Guarded Bayesian validation
Whole sections + neighbour guards
Intervals and simple baselines
Illustrative regional scenario
Uniform bulk-effect multiplier
Positions and zeros retained
Bulk effect only
Solid: analysis flow
Dashed: assumed regional transfer, not an observed cellular response